# INSTITUTE OF SPACE TECHNOLOGY ISLAMABAD



# ELECTRONIC CIRCUITS AND DEVICES

# ECD FINAL LAB PROJECT

## TITLE: 3-INPUT MICROPHONE PRE-AMPLIFIER

## EE-19B


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## OBJECTIVE

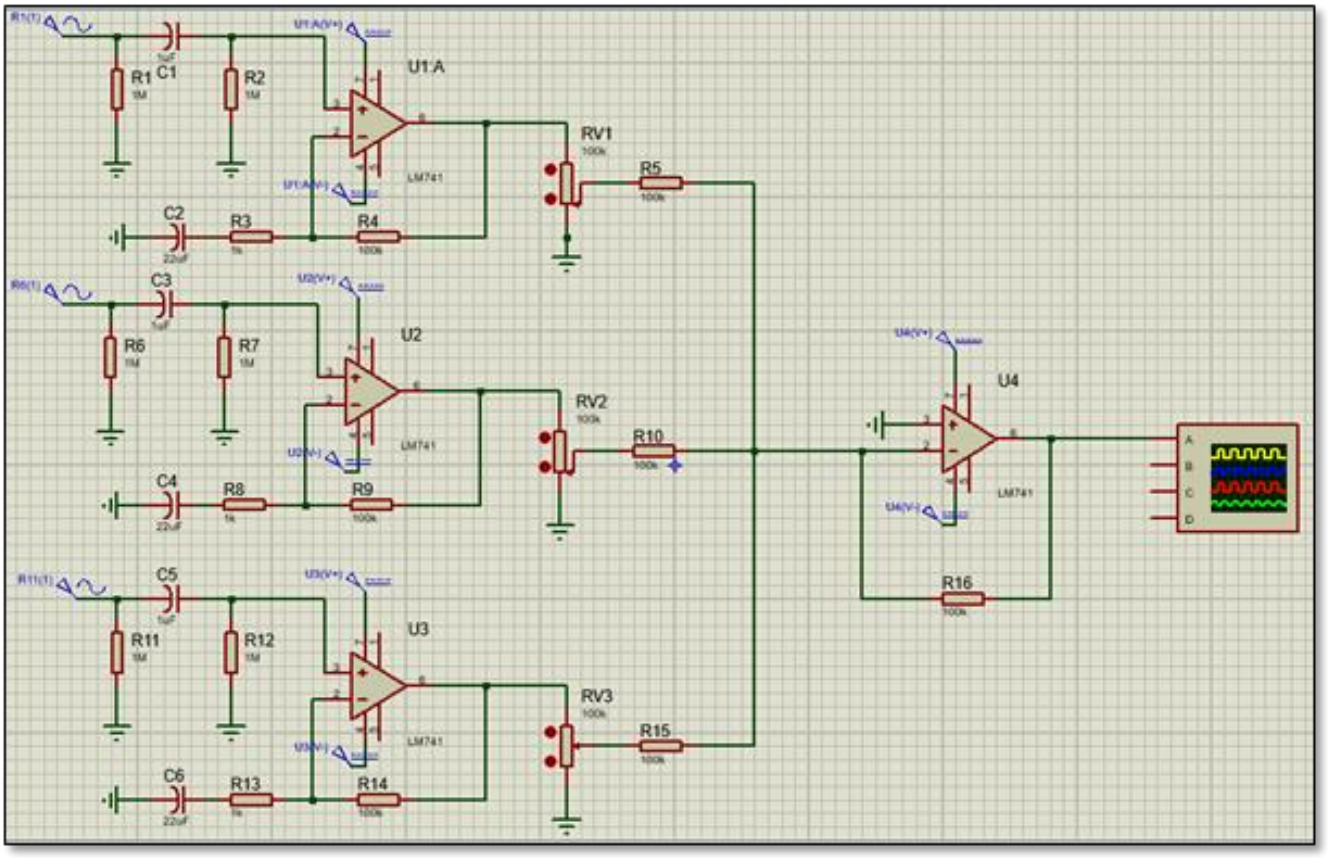
To understand and build a 3-input microphone pre-amplifier in proteus and analyze its output and then implement that design in hardware.

## APPARATUS

- LM 348 (Quad LM741)
- Resistor
- Potentiometer
- Capacitor
- Oscilloscope
- Power supply

## PROCEDURE

- Build following circuit in proteus for simulation and further analysis:




- Set Mic input 1 = 10mV @ 10KHz, Mic input 2 = 10mV @1KHz and Mic input 3 = 10mV @ 100Hz.

$$\text{Gain} = 1 + \frac{\text{Rf}}{\text{Ri}}$$

$$Gain = 1 + \frac{100k}{1k}$$

$$Gain = 1 + 100$$

$$Gain = 101\ V/V$$

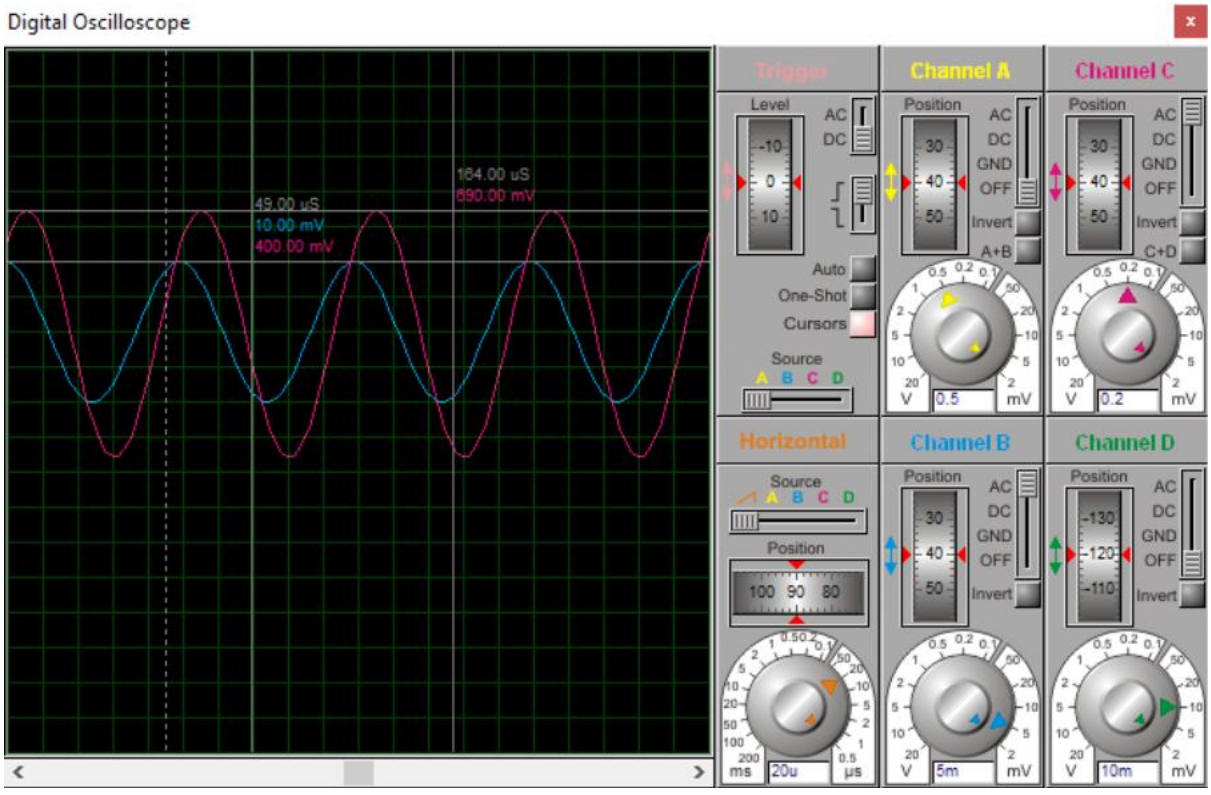
- Set an oscilloscope to analyze the gain of pre-amplifier. Connect any of the output of pre-amplifier and note the following readings:

  ***Vin = 10mV***

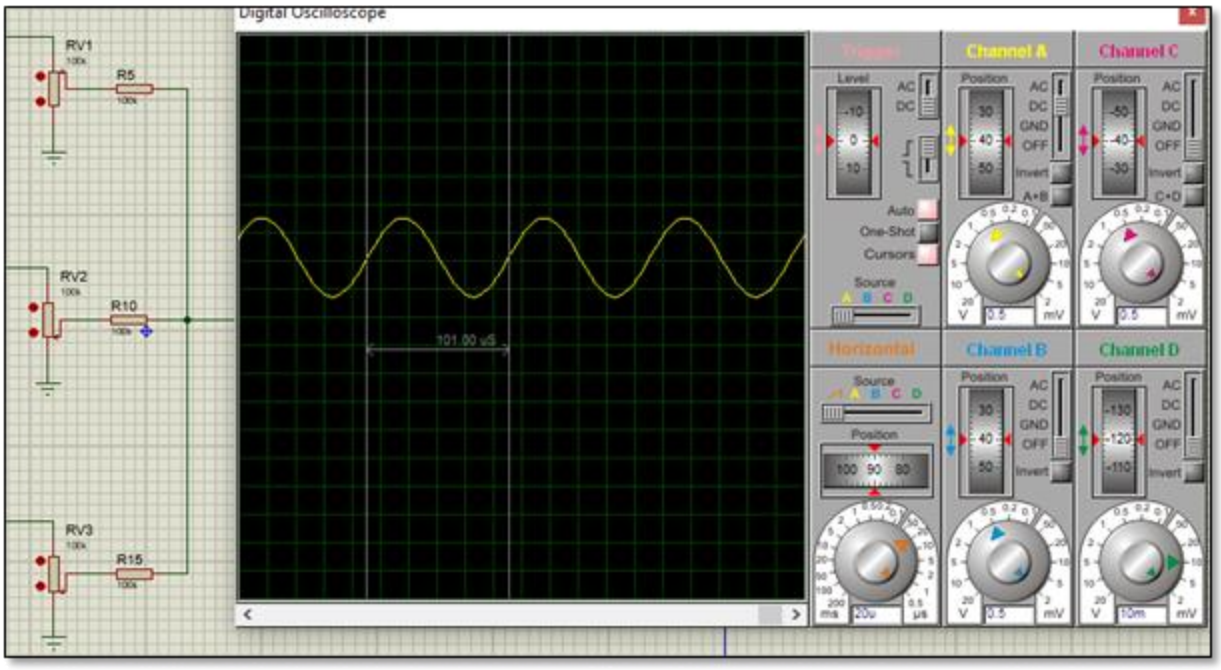
  ***Vout = 0.9V***

  ***Gain = Vout/Vin = 0.9V/10mV***

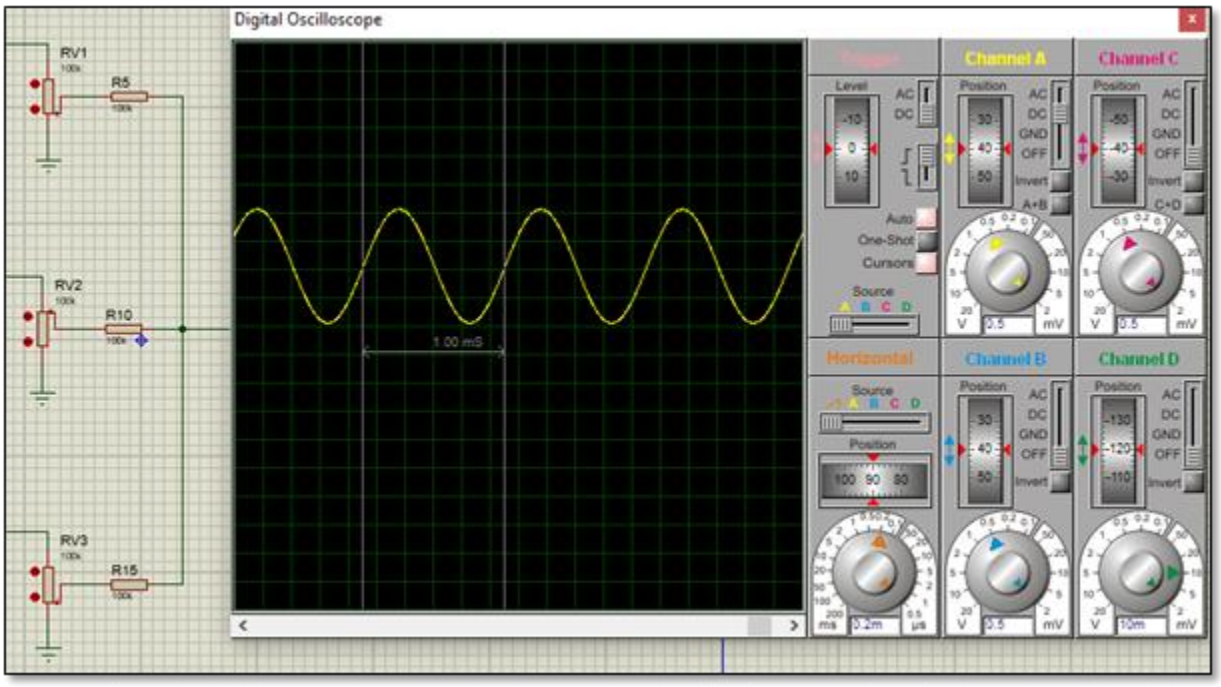
  ***Gain = 90***



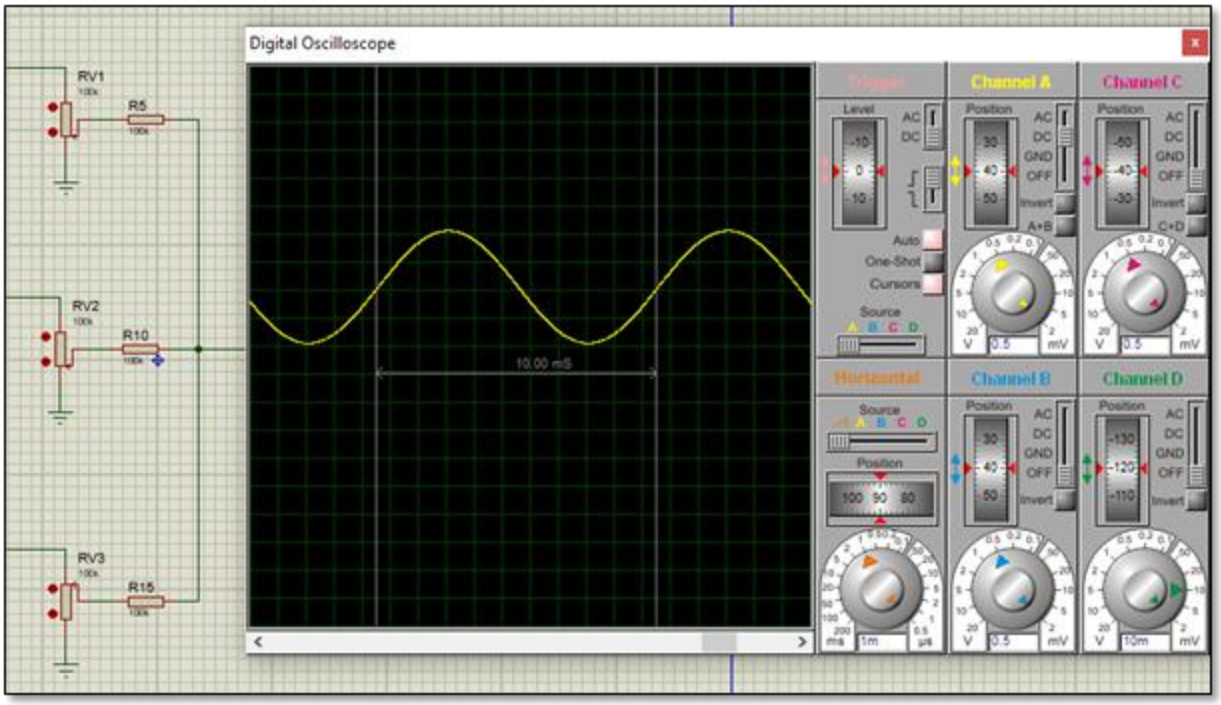

- Set amplitude of input 1 maximum by setting potentiometer to its maximum value. Minimize the amplitude of both other inputs. Analyze the output:



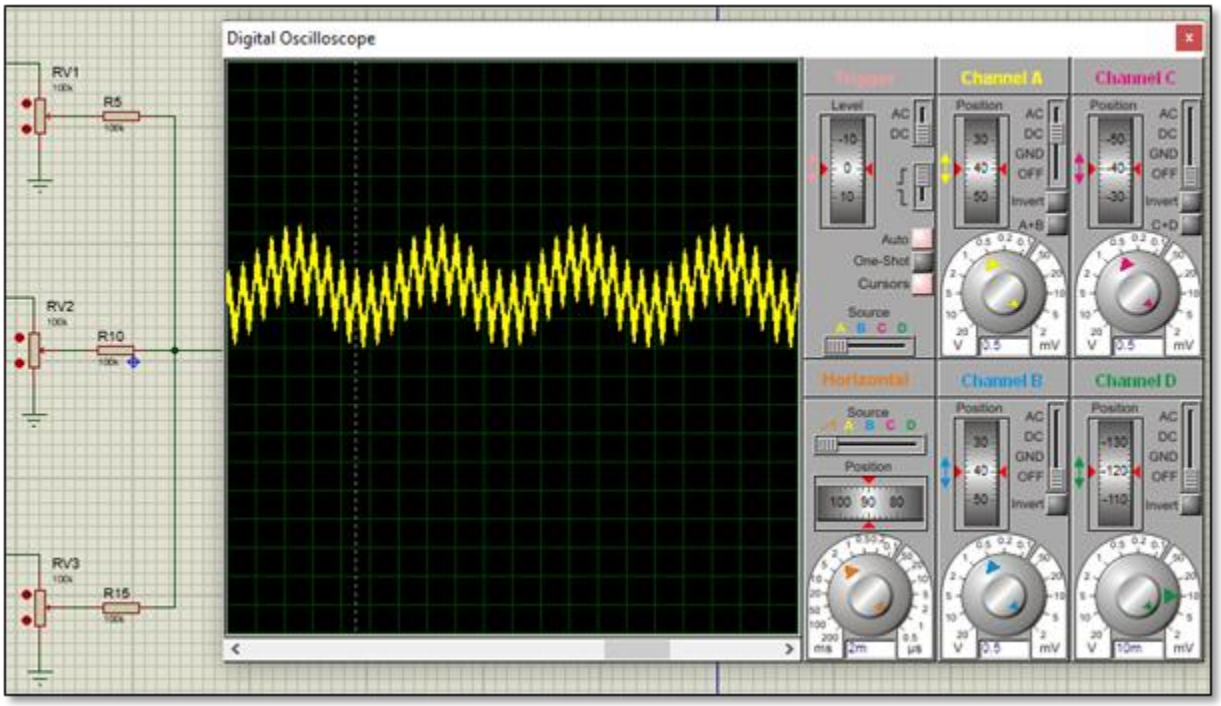

- Now set input 2 to maximum and both other inputs to minimum. Analyze the output:




- Now set input 3 to maximum and both other inputs to minimum. Analyze the output:




- Now set all inputs to 50% of their amplitude and analyze the output:




- Note down the following readings in table:

| Mic In1 | Mic In2 | Mic In3 | Vout | fout | Θout |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| 100 % | 0 % | 0 % | 1 Vpk | 10 KHz | 180 degree |
| 0% | 100 % | 0 % | 1 Vpk | 1 KHz | 180 degree |
| 0 % | 0 % | 100 % | 1 Vpk | 100 Hz | 180 degree |

**CONCLUSION**

Three input pre-amplifier have been implemented and analyzed on proteus and on hardware successfully.